# **XAVIER AYUB** | Selected Works

I am an **urban designer and planner** with a **Master in City Planning** from **MIT**
and a **BSc in Urban Planning, Design and Management** from **The Bartlett, UCL**

Prior to graduate school I spent three years as an **Urban Designer** at **Broadway Malyan**
in London, working across **large-scale masterplanning projects** and **neighborhood design**

My work is guided by a **justice-oriented approach** that centers **communities** and
**human rights** in spatial decision-making

All work shown is **my own** unless stated otherwise

# **01** Urban Design I

Info

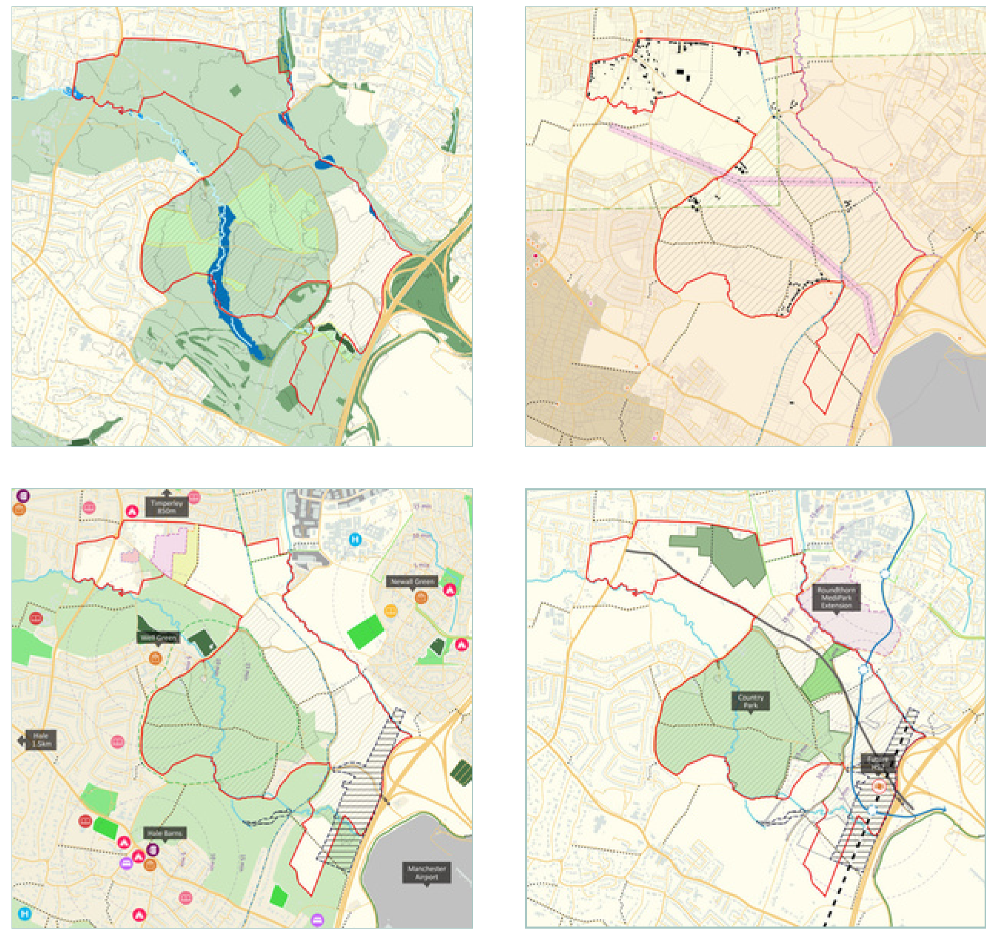
Davenport Green Framework Masterplan 2023 - 2024
Professional Work

Software

- QGIS
- Adobe Suite
- AutoCAD & Civil 3D

Summary

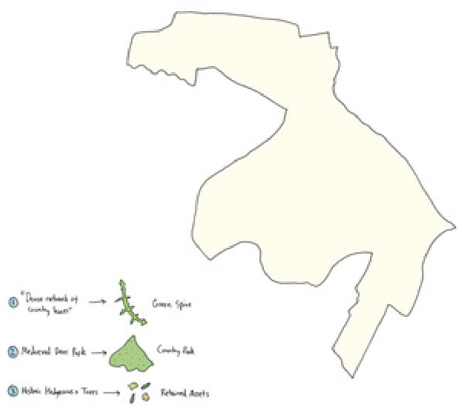
From 2023 to 2024 I worked on a masterplan for a 225 ha site near Manchester Airport. Our design team consisted of myself and two directors from my company's Manchester studio who have been managing the project while I led on site analysis and design. The site is allocated in the Greater Manchester Combined Authority's 'Places for Everyone' long-term plan for Manchester which specifies over 2,500 new homes and 60,000 sqm of employment floorspace for the site. In light of extreme local disparity, with the affluent neighborhood of Hale to the south and the deprived neighborhood of Wythenshawe to the north, the masterplan aims to increase access to affordable homes, jobs, education, green space and public transport while enhancing biodiversity and climate sustainability.



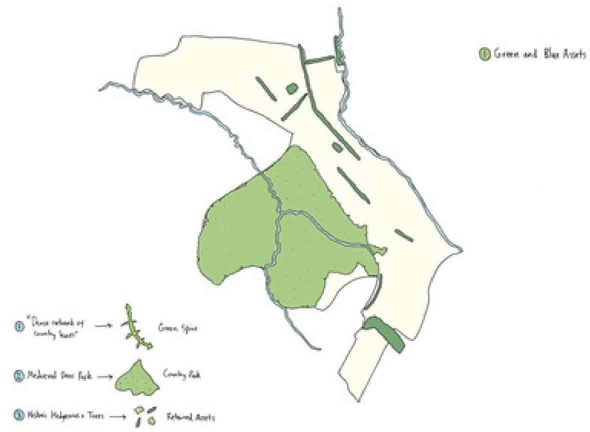
Series of baseline analysis plans that I prepared using QGIS and Adobe Illustrator

Below is a set of rough sketches of **'design driver' diagrams** showing my rationale and design process for the masterplan concept.

As this was intended to be a landscape-led masterplan, I began by identifying **three key environmental elements** of the existing site: a dense network of **tree-lined country lanes**, the site of a former **medieval deer park**, and a series of **historic hedgerows and trees** across the site. These are carried through into the masterplan in the form of the **'green spine'**, a new **country park**, and the retention of **ecologically valuable hedgerows and trees** respectively.



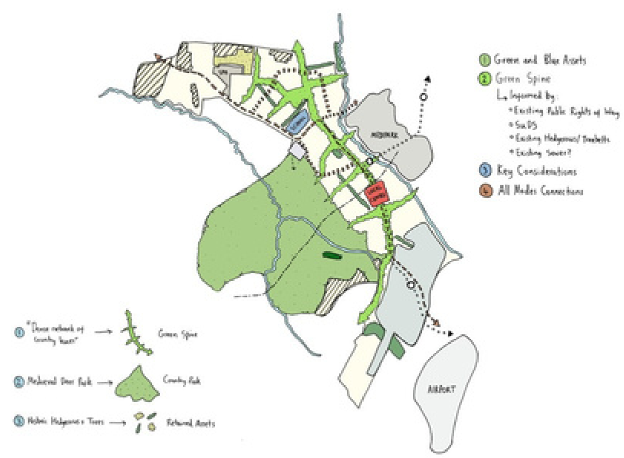
1) Considering the Character of the Site



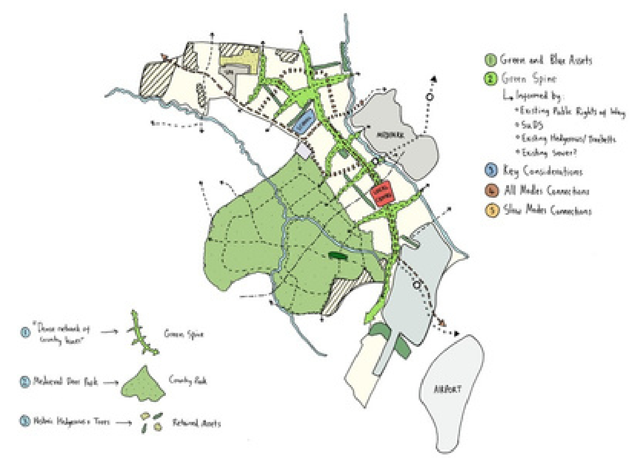
2) Structuring around Green and Blue Assets

3) Connecting with a Green Spine

4) Accounting for Key Constraints and Land Uses



5) Establishing a Movement Network

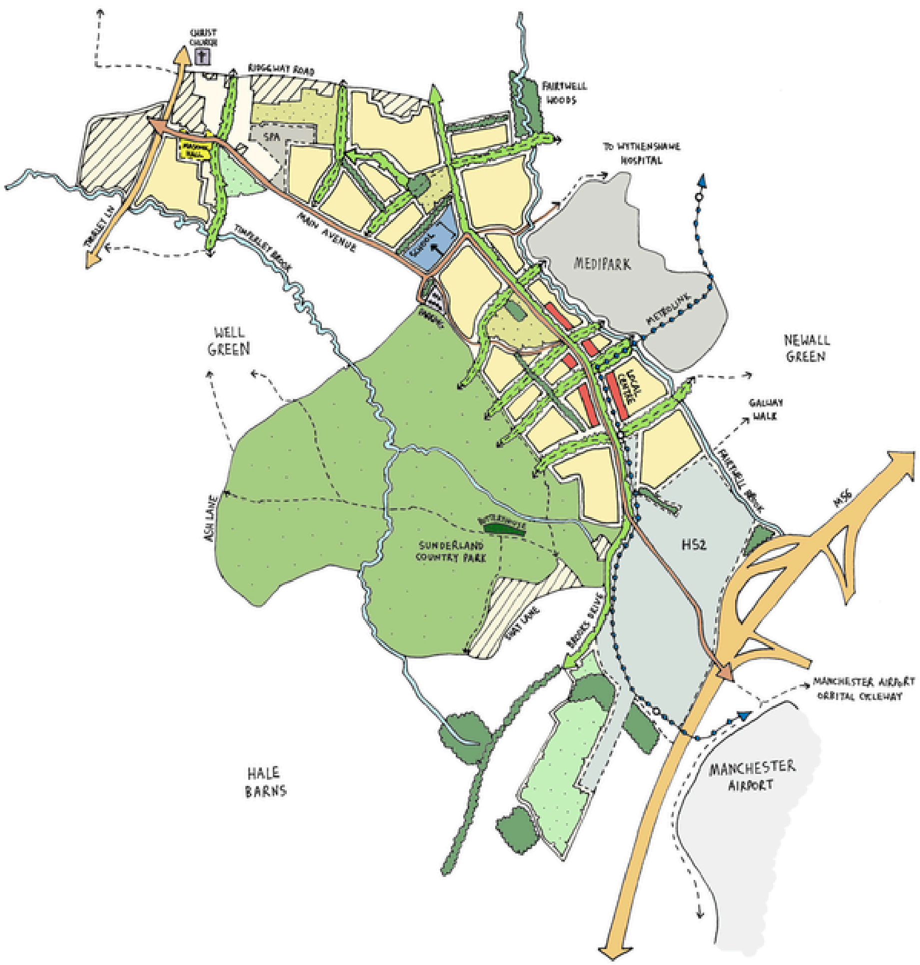


6) Integrating Additional Walking & Cycling Routes

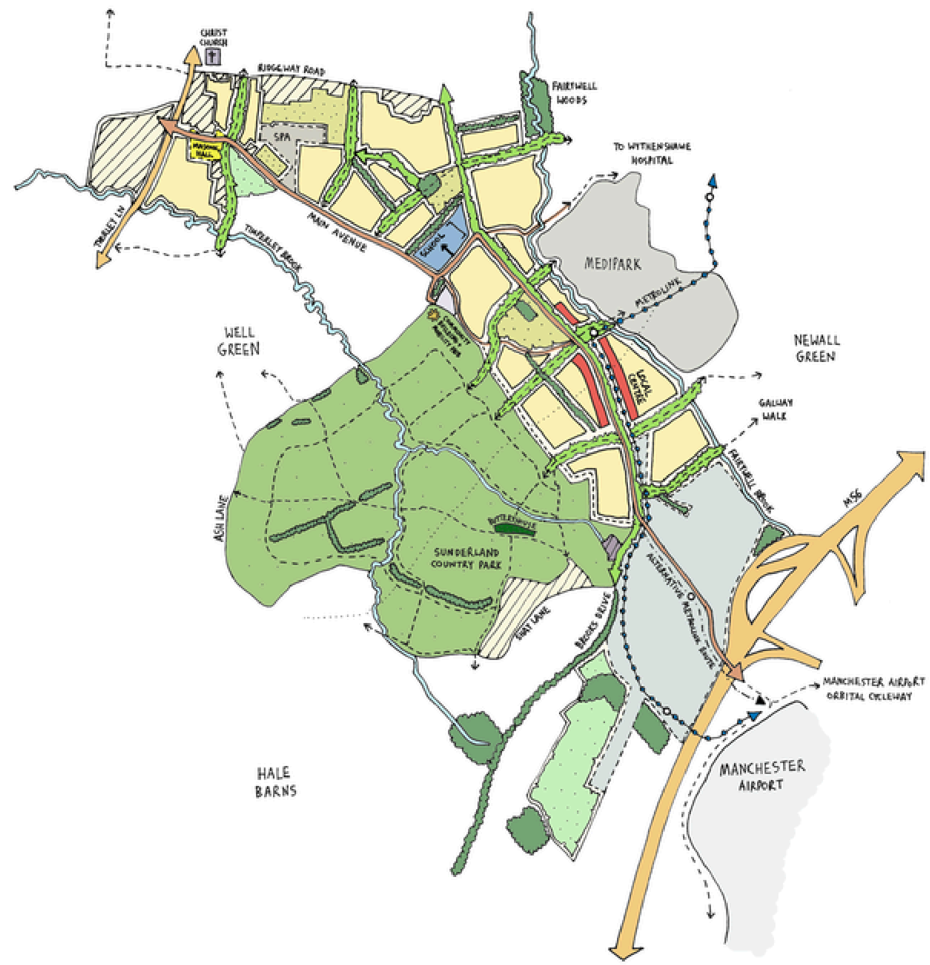
My **initial sketch** (left) builds on the concept of a central **'green spine'** running from north to south, acting as the backbone of the development. This is integrated with east-west **'green spurs'** accommodating active travel and connecting this spine to surrounding land uses. The design retains existing homes and **reduces impact on current residents** by using **green space as buffers**. It also incorporates a **primary school** (blue), **local center** (red), and **metrolink extension**.

Following discussion with my supervisors I drew a **revised sketch** (right) with several changes including the addition of a more detailed network of **walking routes through the country park** and **removal of the HS2 station** following the cancellation of the project by the UK government.

I was then responsible for **optioneering work** on this layout so we could present a series of different layouts for the client and project team to discuss.




Initial sketch of framework masterplan




Revised sketch of framework masterplan

I summarized different location options for key land uses into **four distinct layouts**.

Each of these incorporated a unique option for the following elements:

- **Local Center Location** (red)
- **Primary School Location** (blue)
- **Green Wedge Extent**
- **Metrolink Alignment**

Each layout is either 'Landscape Heavy' or 'Landscape Light' and 'Focused' or 'Distributed'.

'Landscape Heavy' entails an extensive green wedge surrounding the local center.

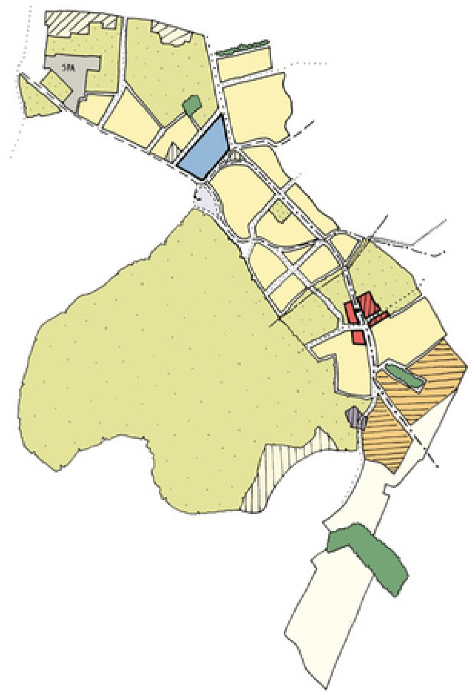
'Landscape Light' entails a minimal green wedge running to the north of the local center.

'Focused' entails key land uses in the form of the primary school and local center being co-located.

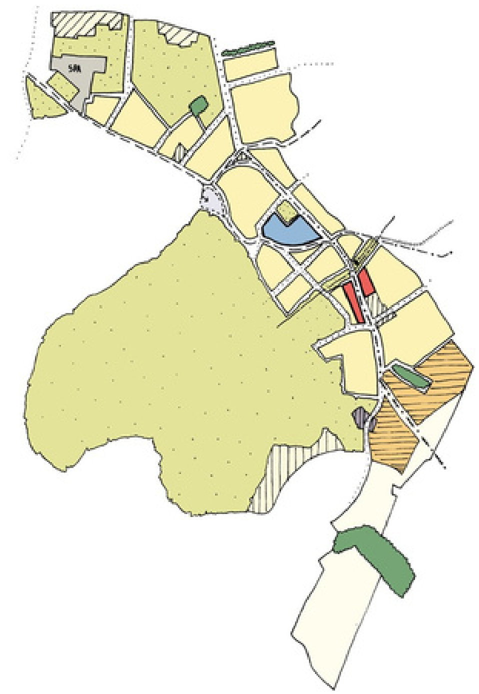
'Distributed' entails key land uses being separated to form two centers: one more community-focused around the school, and one more commercial-focused around the mixed-use local center.

Ultimately we settled on the local center location from the top left diagram, the school location from the top right diagram, and the green wedge extent and metrolink alignment from the bottom left diagram.

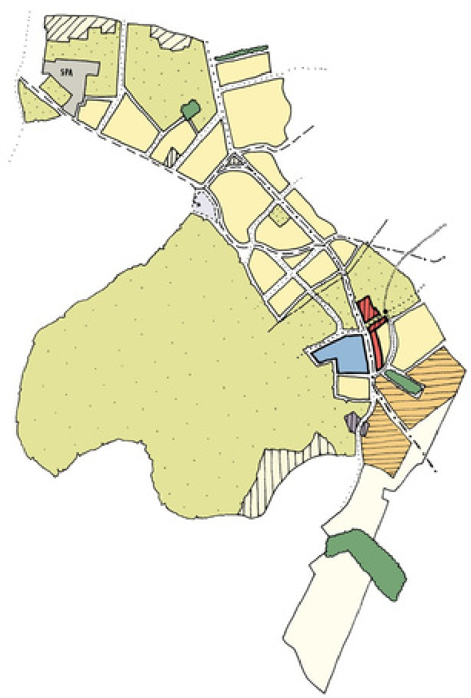
**Distributed & Landscape Heavy**



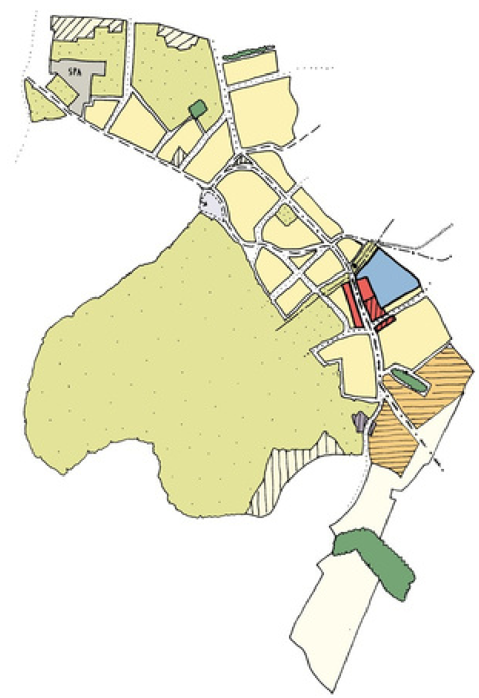
**Distributed & Landscape Light**



**Focused & Landscape Heavy**



**Focused & Landscape Light**




Drawing of the masterplan submitted for public consultation in January 2024


**3D illustrative drawing of masterplan highlighting key land uses submitted for public consultation in January 2024**

# **02** Urban Design II

Info

Nuneaton Town Centre Masterplan

2023

Professional Work

Software

- QGIS
- Adobe Suite
- Affinity Suite

Summary

Over the course of several months I worked with Alejandro de Miguel Solano, a senior colleague in my company's UK Urbanism team, to produce a masterplan on behalf of Nuneaton and Bedworth Borough Council. This entailed the regeneration of the train station and town center of a market town called Nuneaton. I began by undertaking a baseline analysis of the area using QGIS and produced a series of plans using Adobe Illustrator. Following a workshop with local stakeholders we produced a SWOT analysis for the town center and formulated a set of principles to guide our masterplan. I came up with an initial concept for the masterplan which we then refined. Key focuses were increasing public realm safety and connectivity, providing new open space, and activating the waterfront of the town's largely neglected river.

**Define a Strong Identity**
- Establish clear character areas
- Shift perceptions
- Create a new station neighborhood

**Create a Station Gateway**
- New platform/station experience
- Station presence improved
- New Station Square and Anker River Park

**Unlock the River**
- Pedestrian and cycling routes along river and associated green spaces with new Anker River Park
- Residential development and public activities along riverfront

**Seed a Local Community**
- Include residential and supporting social infrastructure
- Consider most suitable anchor uses to generate buzz

**Mix it Up**
- Range of uses to increase dwell times
- Foster evening economy by extending business operating hours

**Identify Development Opportunities**
- Identify short, medium and longterm development opportunities
- Identify quick wins, funding & delivery mechanisms
- Consider how increased accessibility boosts value

**Reduce Car Dominance & Severance**
- Downgrade ringroad
- Consolidate parking
- Remove private vehicle access (taxi/bus pick-up only to south)

**Reconnect North & South**
- Pedestrian/cyclebridgeoverrailway connecting existing & new communities
- New pick-up & drop-off
- New travel hub to the north of station

**Join the Dots**
- Weave nodes/attractors (e.g. colleges & MIRA) into a sustainable transport network
- Utilise river and existing greenways to connect local communities to town center, station and social infrastructure

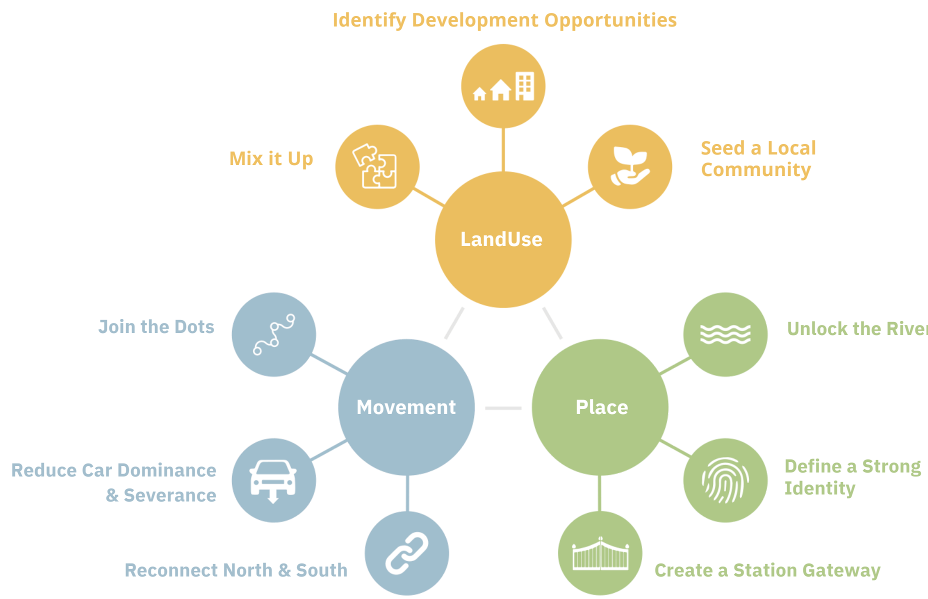
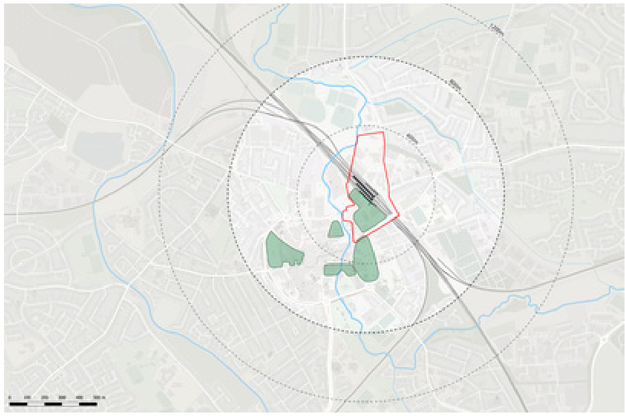
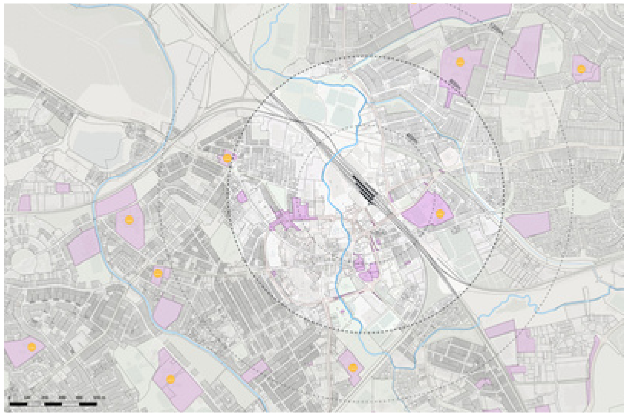
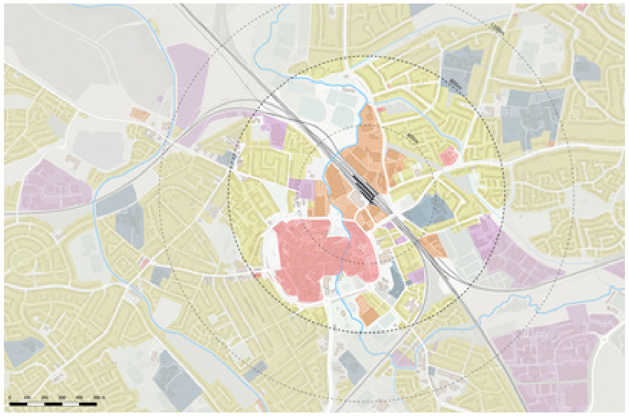
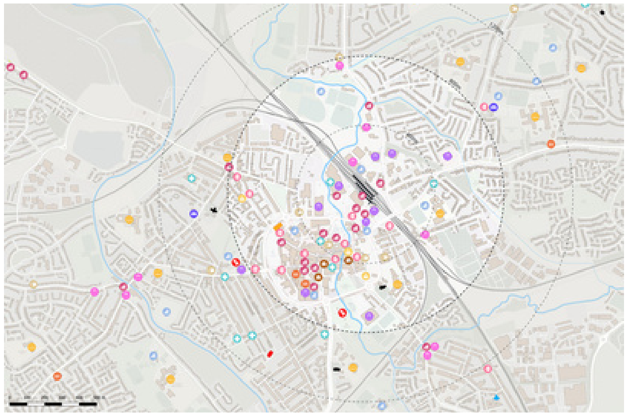


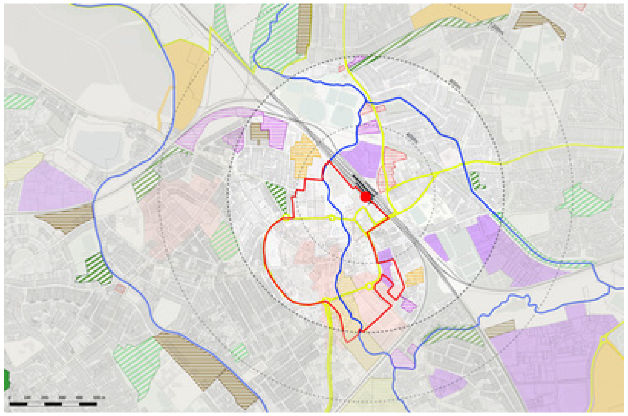

Diagram showing the principles for the masterplan based on feedback from local residents and other stakeholders

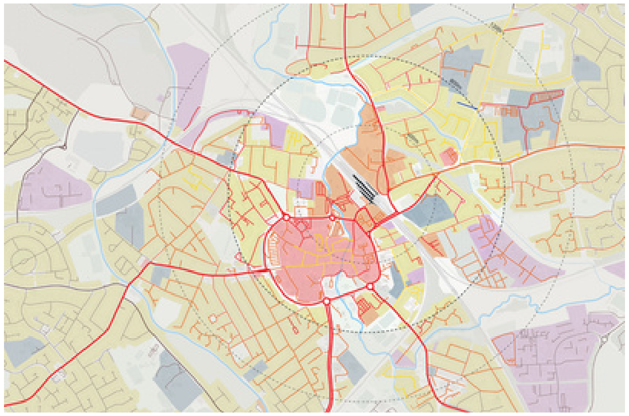

Regeneration Projects

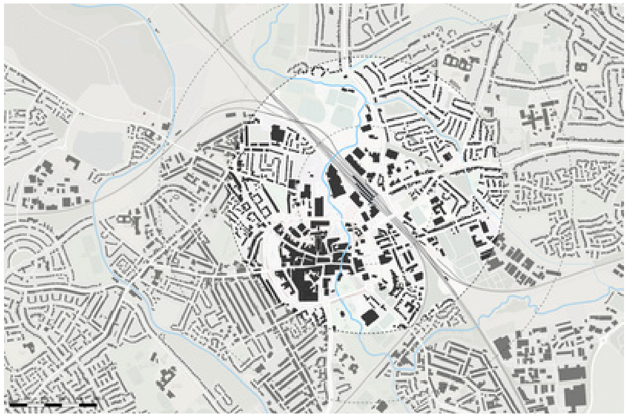

Land Ownership / Assembly

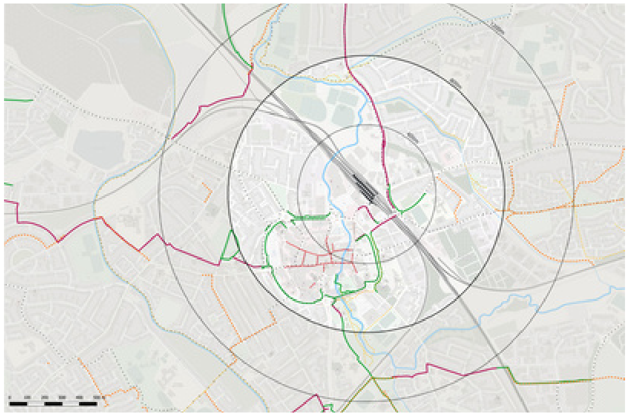

Land Use

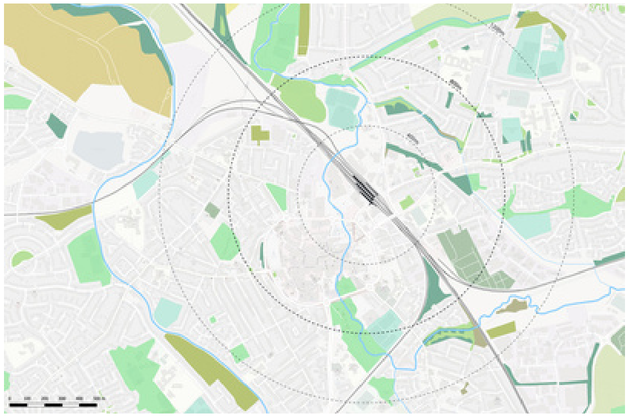

Local Facilities

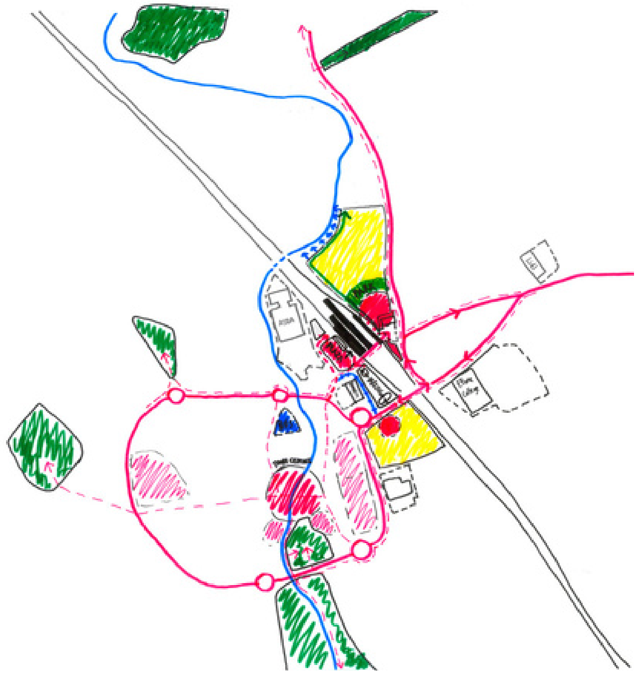

Planning Policy


Public Realm Comfort & Safety


Urban Grain

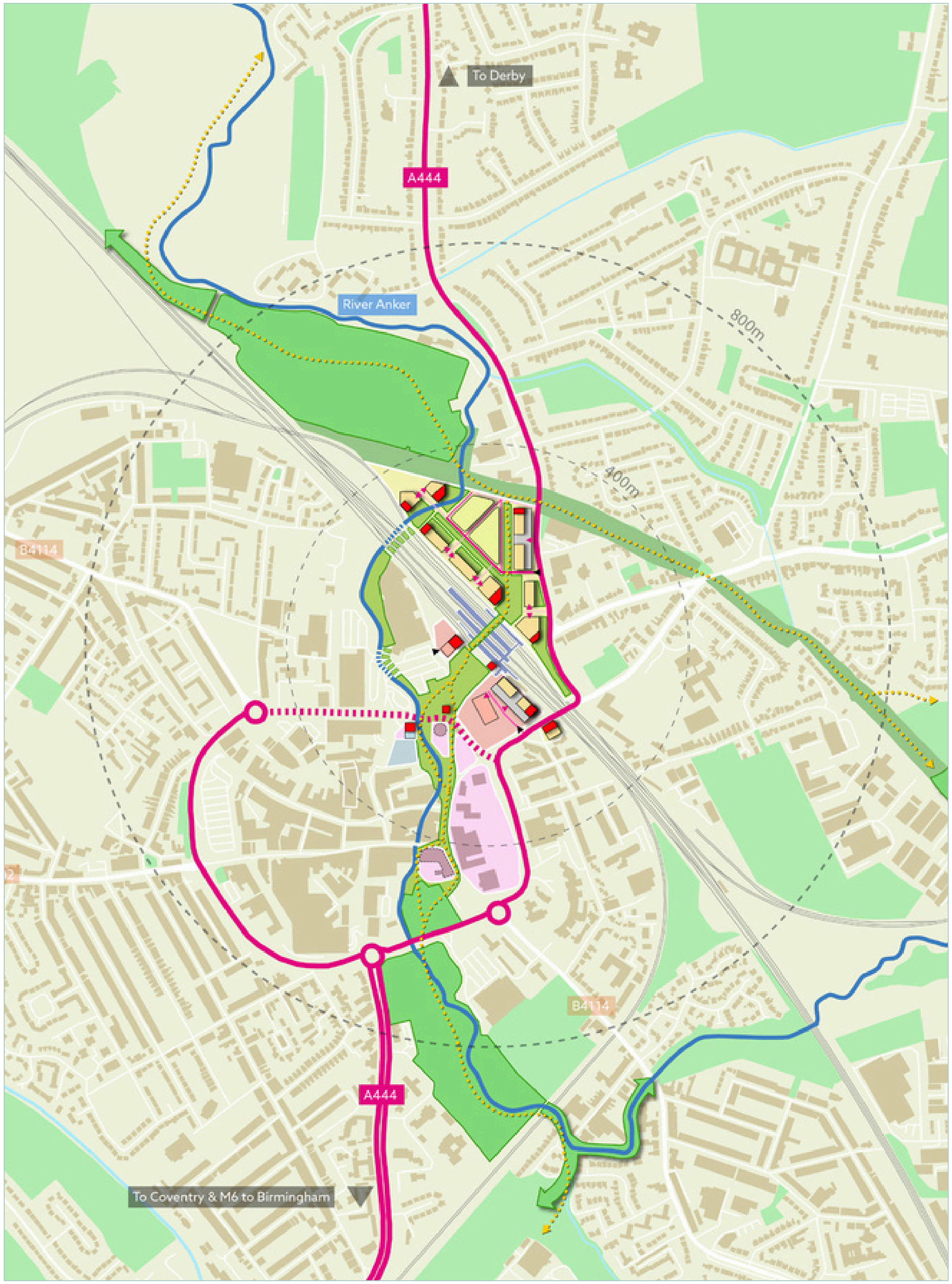

Pedestrian & Cycle Mobility


Green and Blue Infrastructure

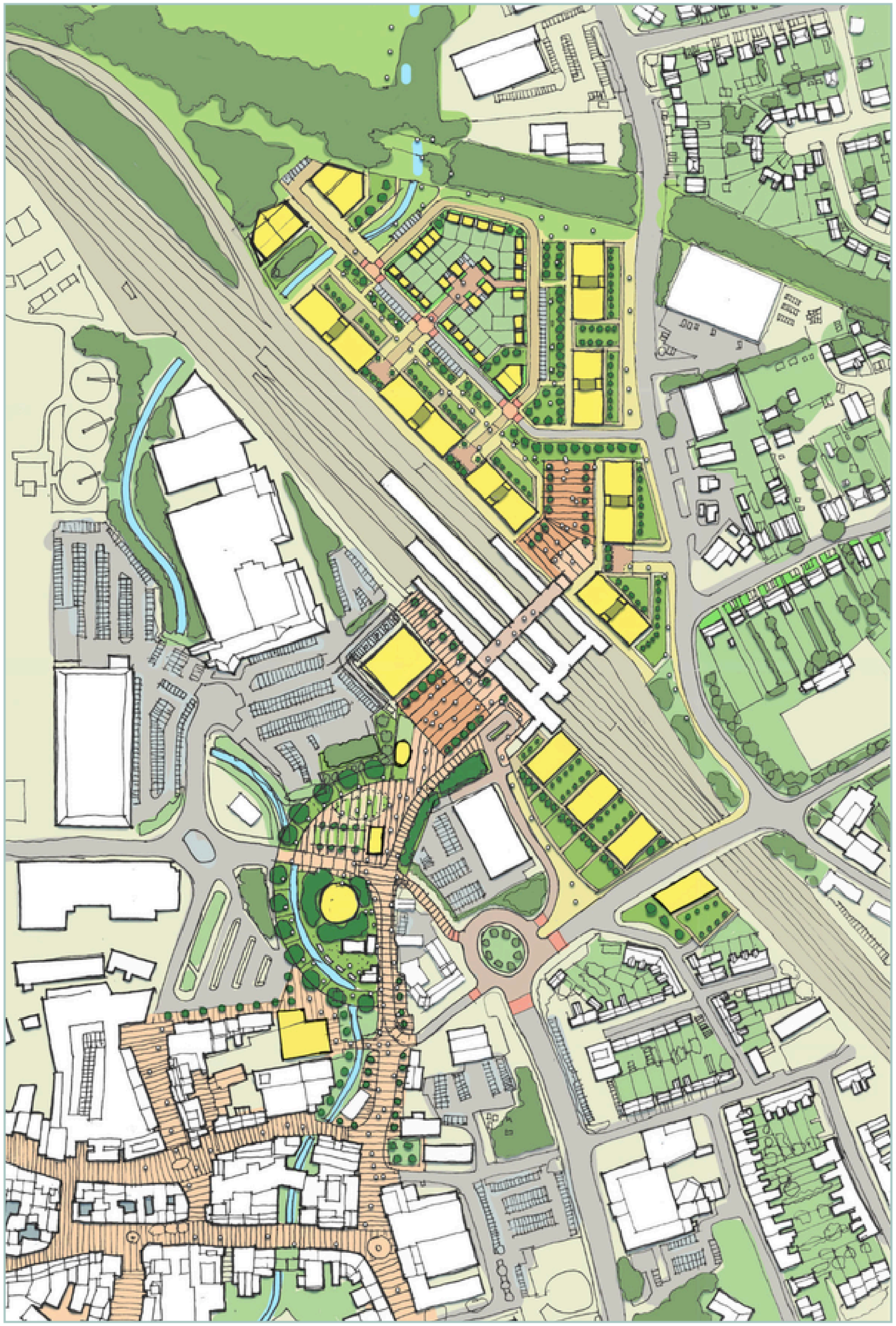


Rough Sketch of my Initial Ideas for the Strategic Concept

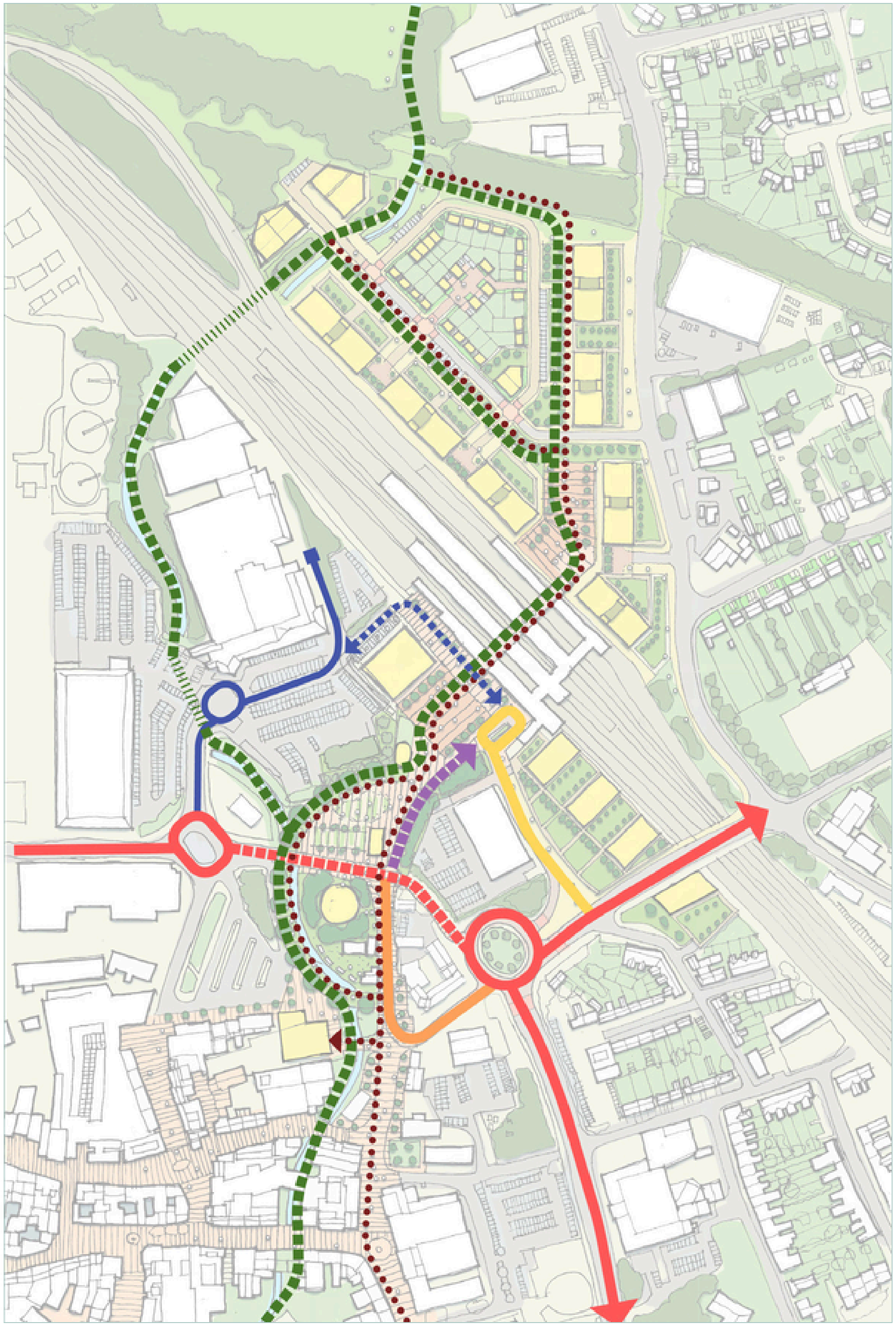
Rough Sketch of my Initial Ideas for the Station Regeneration




Initial Ideas translated into a Strategic Concept Plan



Final Masterplan (drawing by Alejandro de Miguel Solano)



Final Movement Plan (drawing by Alejandro de Miguel Solano)

# **03** Spatial Analysis

Info

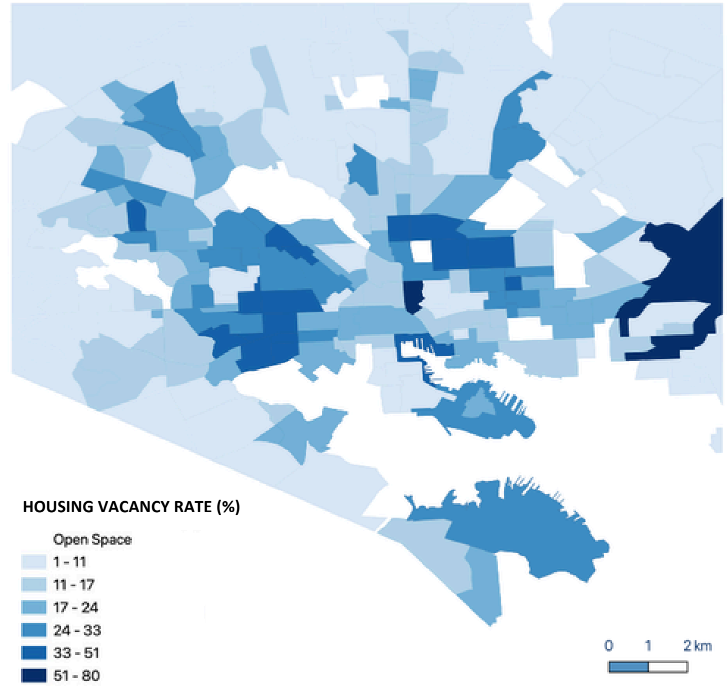
Vacant Housing and Crime in Baltimore
2019
University Project (UCL)

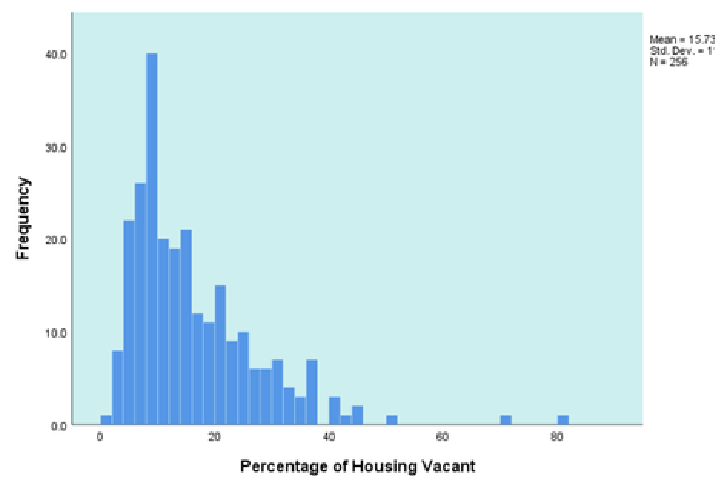
Skills

- QGIS
- SPSS Statistics
- Excel

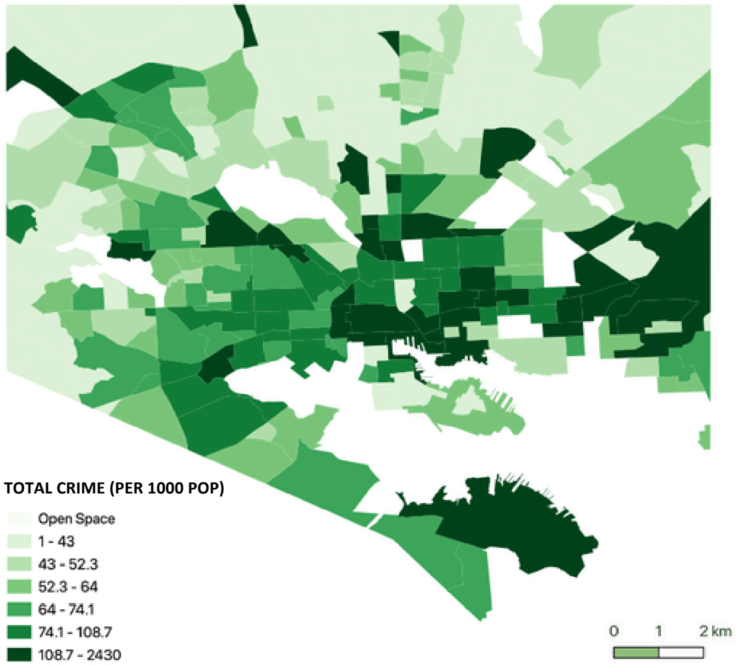
Summary

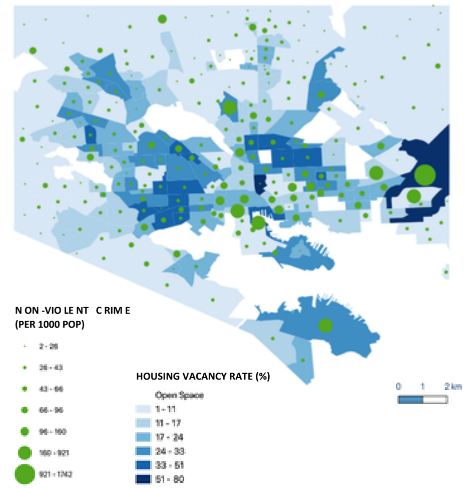
This spatial analysis project examined the relationship between vacant housing rates and crime in Baltimore, Maryland. Though conceptually prompted by the controversial 'broken windows' theory (often used to justify aggressive policing tactics like 'stop-and-frisk'), the project sought to contextualize the theory's assumptions within Baltimore's specific history of racial and economic inequality. My visualizations and correlation analysis confirmed only a weak positive relationship, undermining a core premise of the theory. Crucially, they illuminated how this correlation is not causal but is fundamentally driven by a key confounding variable: the city's entrenched geography of segregation, visually captured by Baltimore's distinct 'Black Butterfly', which is a direct legacy of historical redlining and ongoing disinvestment. The findings suggested that crime and vacancy are co-occurring outcomes of this long-term systemic neglect, not factors in a simple causal chain.



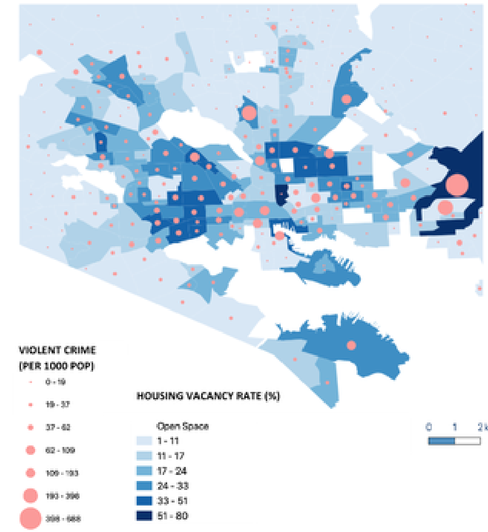




Map and histogram of housing vacancy rates in Baltimore neighborhoods

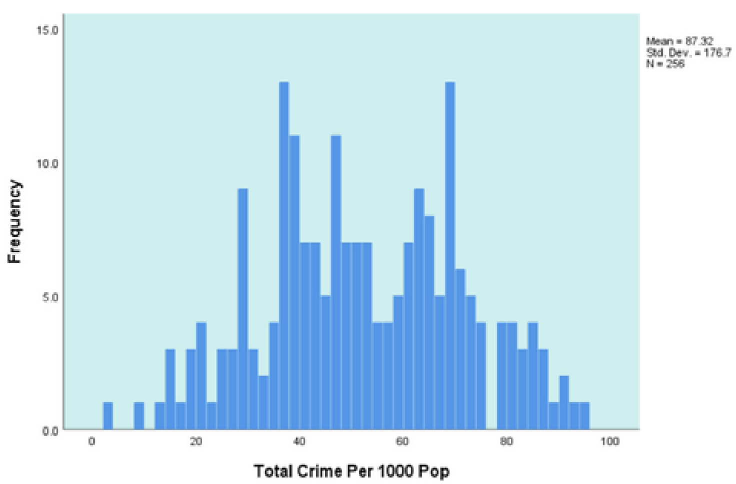


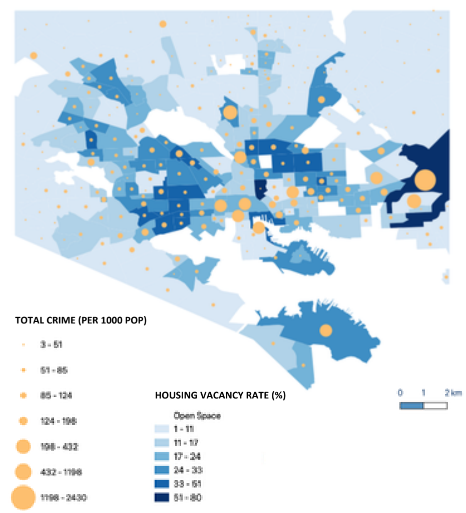

Map showing vacancy rates in Baltimore compared with non-violent crime rates



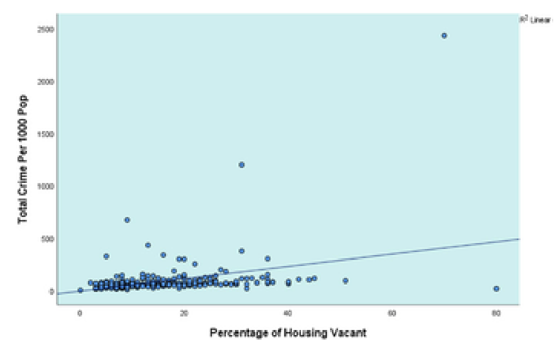

Map showing vacancy rates in Baltimore compared with violent crime rates




Map and histogram of crime rates in Baltimore neighborhoods




Map showing vacancy rates in Baltimore compared with total crime rates




Scatterplot showing the relationship between crime rate and housing vacancy rate in Baltimore neighborhoods

# **04** Urban Data Science

Info

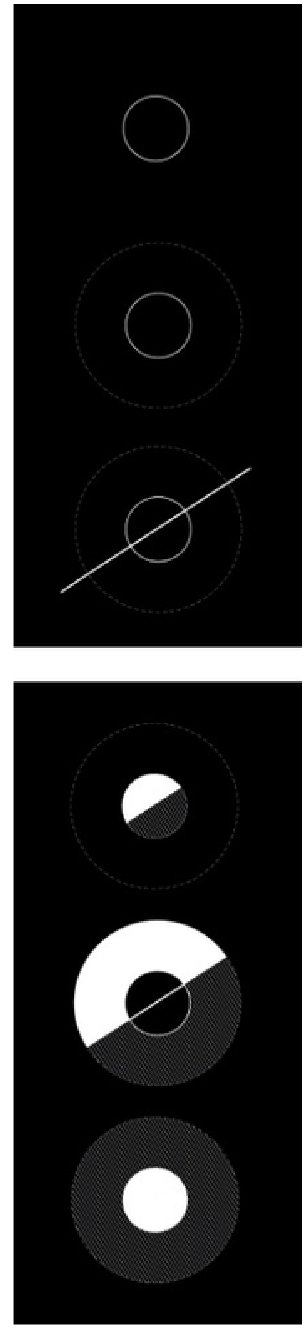
Programming Project on Nicosia ([Link](Link))
2025
Group University Project (MIT)

Skills

- Programming using R
- Data Visualization
- Interactive Web Application Development

Summary

I conceptualized and led a quantitative urban analysis of Nicosia, the divided capital of Cyprus, working in a five-person team to produce an interactive web app and analytical report. To address political fragmentation and data scarcity, we developed a novel spatial framework that manually delineated a “Greater Nicosia” urban extent using satellite imagery to enable cross-border comparability. The analysis employed OSM-derived metrics (including building density, road networks, places of worship, and commercial activity) as proxies for urbanization, ethno-religious division, and economic disparity across three contexts: within the historic Venetian walls, outside the walls, and across the UN Buffer Zone. The work revealed disparities between the north (TRNC) and south (RoC) and demonstrated how computational spatial analysis can generate insight in politically complex, data-poor environments.



Historic Walled City of Nicosia

Walled City within Greater Nicosia

Nicosia divided by UN Buffer Zone

1) Inside Walls: RoC vs. TRNC

2) Outside Walls: RoC vs. TRNC

3) Inside Walls: RoC vs. TRNC

Conceptual diagrams showing the approach to comparison taken in analyzing Nicosia

1) Inside Walls: RoC vs. TRNC | 2) Outside Walls: RoC vs. TRNC | 3) Inside Walls vs. Outside Walls

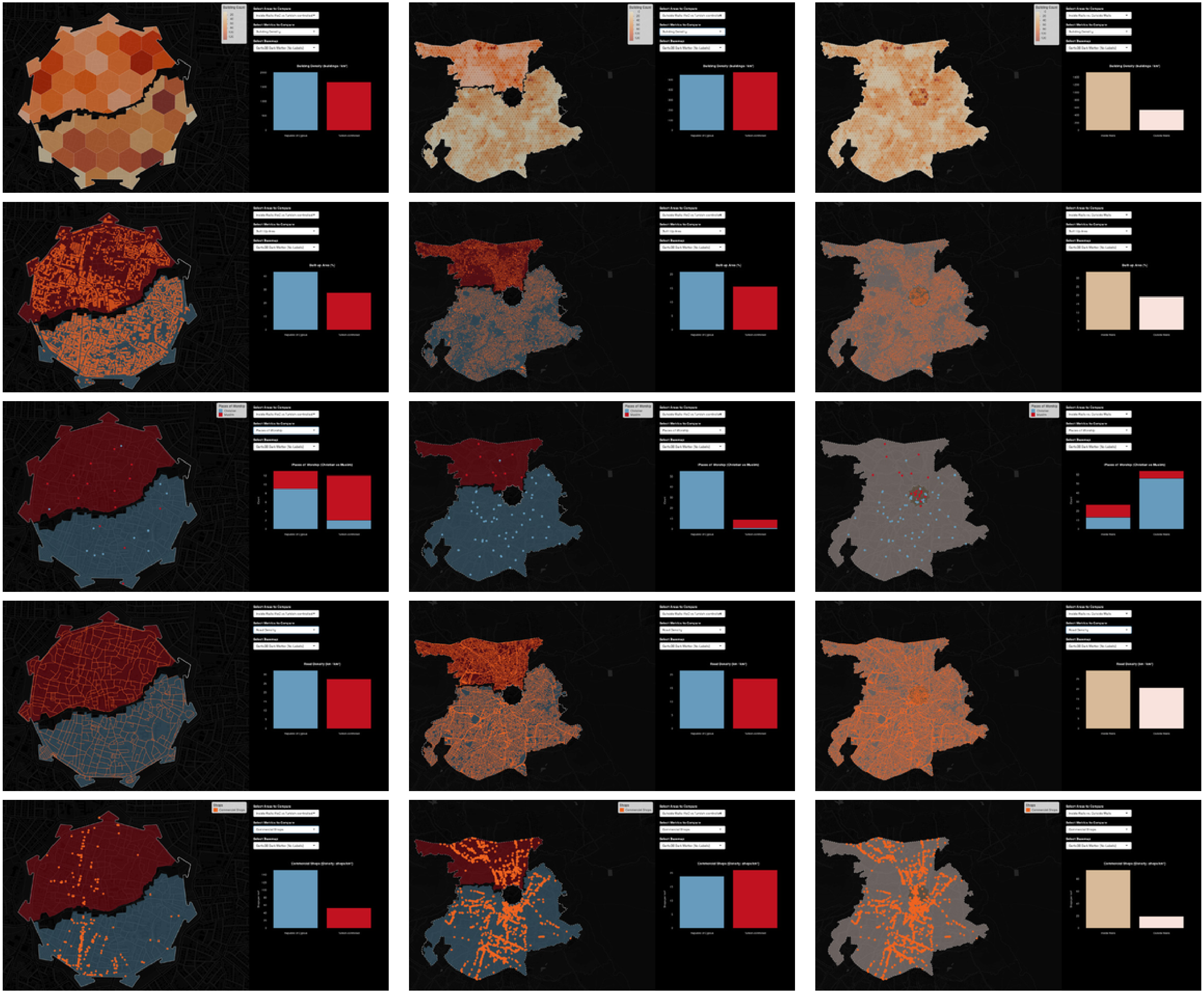
Building Density

Built Up Area

Places of Worship

Road Density

Commercial Shops




**XAVIER AYUB** | End of Portfolio